RÉPUBLIQUE DU CAMEROUN
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MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR
....
COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN HIGHER NATIONAL DIPLOMA EXAM (HND)
....

REPUBLIC OF CAMEROON
Peace-Work-Fatherland
....
MINISTRY OF HIGHER EDUCATION
....
NATIONAL COMMISSION FOR THE ORGANISATION OF HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM
....

National Exam of Higher National Diploma 2022 Session

Spécialty/option: SWE, CSN, DBM
Paper : Digital Electronics **Credits : 7/7/4**
Duration : 4 hours

Instructions: *Answer all questions. Calculators of any type are not authorized!*

SECTION A: NUMBER SYSTEM AND CODES (20marks).

MCQs, for each question, four answers are proposed. Write the letter for the correct answer against the question number. Each question carries 1mark.

1) The value of radix in binary number system is ____________

a) 2 b) 8 c) 10 d) 1

2) If the decimal number is made of an integer part and fractional part. Then its binary equivalent is obtained by ________ the fractional part of number continuously by 2.

a) Dividing
b) Multiplying
c) Adding
d) Subtracting

3) An unweight code in which only one bit changes from one code number to the next is __.

a) BCD b) Excess-3 c) Gray d) ASCII

4) The largest BCD number that can be represented with four binary bits is ________.

a) 9 b)10 c)15 d) 16

5) Considering x is a Boolean variable, which of the followings is NOT ALWAYS TRUE?

a) $x + x = x$ b) $x . x = x$ c) $x . 1 = x$ d) $x + 1 = x$

6) The universal gate is ....

a) NAND gate b) OR gate c) AND gate d) None of the above

7) The inverter is ....

a) OR gate b) NOT gate c) AND gate d) None of the above

8) Which of the following logical operations is represented by the + sign in Boolean algebra?

a) Inversion b) AND c) OR d) complementation

9) The binary equivalent of gray code 1111 is :

a) $(1101)_2$ b) $(1011)_2$ c) $(1110)_2$ d) $(1010)_2$

10) The BCD number for decimal 347 is ________.

a) 1100 1011 1000 b) 0011 0100 0111 c) 0011 0100 0001 d) 1100 1011 0110

11) Convert binary to octal: $(110110001010)_2$ =?

a) $(5512)_8$ b) $(6612)_8$ c) $(4532)_8$ d) $(6745)_8$

12) What is the addition of the binary numbers 11011011010 and 010100101?

a) 0111001000 b) 1100110110 c) 11101111111 d) 10011010011

13) Binary subtraction of 100101 – 011110 is

a) 000111 b) 111000 c) 010101 d) 101010

14) Perform multiplication of the binary numbers: 01001 × 01011

a) 001100011 b) 110011100 c) 010100110 d) 101010111

15) Divide the binary numbers: 1010001 ÷ 1000

a) 1010.001 b) 1000.001 c) 1100.001 d) 0011.001

16) Which numbering system uses numbers and letters as symbol?

a) Decimal b) binary c) Octal d) hexadecimal

17) The octal number (651.124)8 is equivalent to ______

a) $(1A9.2A)_{16}$ b) $(1B0.10)_{16}$ c) $(1A8.A3)_{16}$ d) $(1B0.B0)_{16}$

18) The sign of a negative binary number is given by by its ________

a) MSB b) LSB c) Byte d) Nibble

19) The largest two digit hexadecimal number is ________

a) $(FE)_{16}$ b) $(FD)_{16}$ c) $(FF)_{16}$ d) $(EF)_{16}$

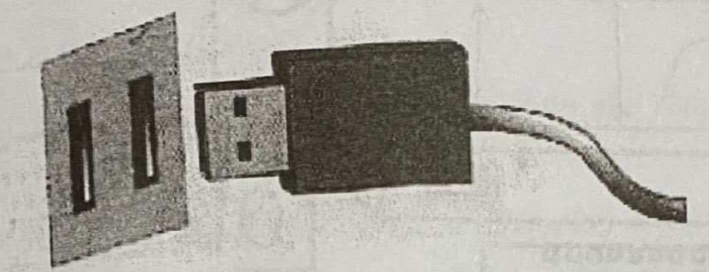
20) The 1 in the binary number 1000 has a place value of --------- in decimal

a) 1000 b) 16 c) -1 d) 8

SECTION B: COMPUTER FUNDAMENTAL (80MKS)

I) HARDWARE (20marks)

1) Study the diagram below and answer questions that follow:



1. a) Identify the port shown in the diagram. **(1mark)**

1. b) Give TWO reasons why the identified port in the figure above finds widespread use in computer systems. **(2marks)**

2) Name the following hardware components and give their role in a computer. ***(7.5marks)***

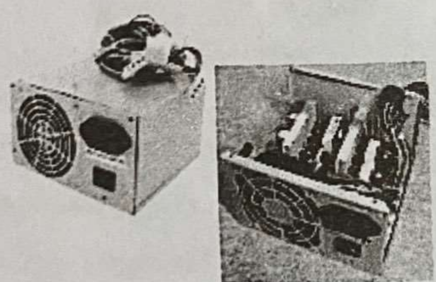


Figure 1

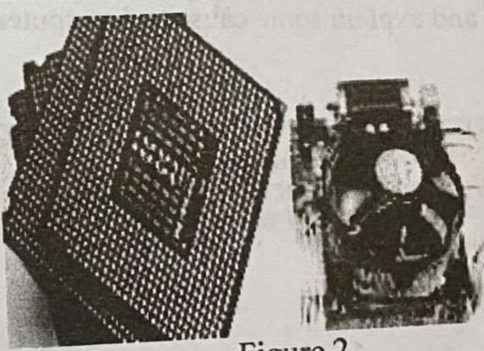


Figure 2

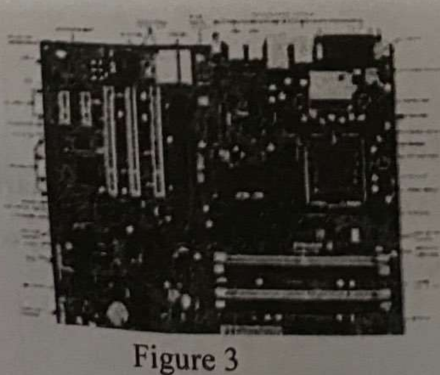


Figure 3

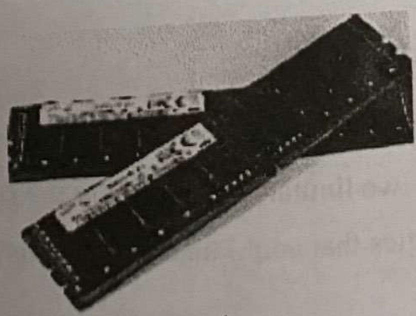


Figure 4

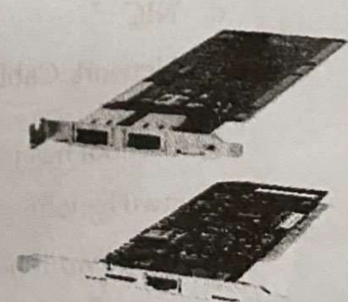
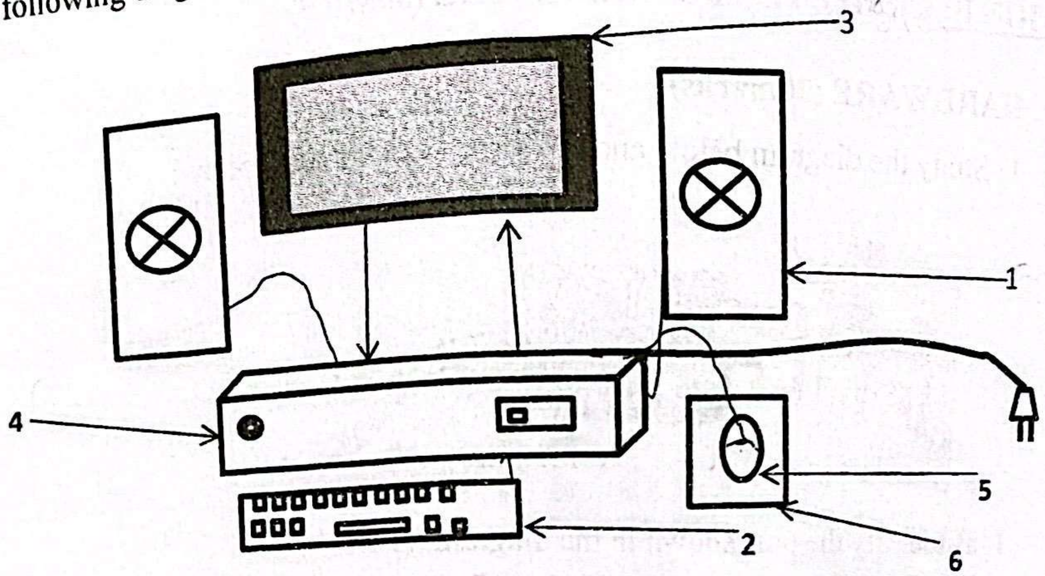


Figure 5

3) What is the main difference between a mainframe computer and a supercomputer? (2.5marks)

4) Given the following diagram below



4. a) Identify the various components from 1 to 6. **(3marks)**

4. b) Identify two input devices and two output devices. **(2marks)**

5) State and explain some causes of computer poor performance. **(2marks)**

II) NETWORK AND MOBILE DEVICES (40MRKS)

1. What are the functions of the following network devices **(5marks)**

a. Distributor

b. Modem

c. Router

d. NIC

e. Network Cable

2. A school has just decided to link its computers on a network.

a) Give two benefits and two limitations of linking the school's computers on a network. **(4marks)**

b) State any two topologies that could be used for this networking, illustrating your answer with diagrams. **(6marks)**

3. a) What is a mobile device? **(2marks)**

b) List two types of mobile devices you know **(3marks)**

c) What are the common types of operating systems used in mobile devices you know **(2marks)**

d) Give three examples of common applications used in mobile devices **(3marks)**

e) Briefly describe in point form one method you will use to connect your mobile phone to a computer for file sharing. **(5marks)**

4. Considering that a company needs 60 usable host ip addresses for its connectivity to the internet. Having a network address of 198.100.10.0. Answer the following questions.

a) What is the address class? **(2marks)**

b) What is the default subnet mask of the above network? **(2marks)**

c) What is the custom subnet mask? **(2marks)**

d) What is the total number of subnets form? **(2marks)**

e) What is the total number host per subnet? **(2marks)**

III) MICROSOFT WORD, EXCEL AND POWERPOINT (20MARKS)

1. Define and give an example of the following. **(4marks)**
   - i) *Spreadsheet software*
   - ii) Presentation Software
   - iii) Microsoft Access
   - iv) Microsoft word
2. A shopkeeper recently transferred his daily records from a ledger to a spreadsheet program.

a) State and explain four advantages that a spreadsheet program has over the recording in a ledger. **(4marks)**

Below is a sample spreadsheet used by the shopkeeper above.

S/N	Item	Quantity	Unit Price	Unit Sell Price	Total Profit	Remark
1	Mambo	25	500	600		Good Business
2	Bread	23	800	1000		Good Business
3	Savon	14	550	700		Good Business
4	Sugar	7	900	1500		Good Business
5	Ground nut Oil	10	1200	1300		Good Business
6	Milk	5	1500	1400		Bad Business
7	Biscuit	9	2500	2600		Good Business
8	Sweet	11	350	300		Bad Business
9	Butter	25	480	500		Good Business
10	Rubbing Oil	85	950	1000		Good Business
				Grand Profit		

b) State the formula that can be used to calculate the total profit for the item mambo in cell G4. **(3marks)**

c) Describe how the formula in G4 can be copied in cells G5 to G13. **(3marks)**

d) Give the formula in G15 that can be used to generate the grand profit. **(3marks)**

e) Given that the quantity of milk and its unit price are changed to 10 and 1000 respectively, state the changes that will be observed on the spreadsheet. **(3marks)**